Boundary-Localised Spectral Supervision for Image Inpainting: Fixed vs. Adaptive Band Weighting

Anonymous Authors
Anonymous Institution
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***Abstract*—Image inpainting methods frequently produce visually plausible hole fills yet exhibit discernible artefacts at the boundary seam between the inpainted and known regions. Existing loss formulations apply global reconstruction objectives and, at best, uniform spectral penalties that ignore the frequency-dependent nature of seam visibility. We present a novel *Adaptive Spectral Boundary Coherence* (ASBC) loss that decomposes the discrete Fourier transform of localized boundary patches into N radial frequency bands and learns per-band importance weights end-to-end through a compact set of parameters. An entropy regularizer prevents weight collapse, ensuring that all frequency bands contribute meaningfully during training. We evaluate ASBC within a boundary-aware encoder–decoder implementation augmented with a Transformer bottleneck, mask-injected skip gates, and boundary-conditioning FiLM modules. Ablation over six loss configurations on Places365 shows that naive fixed-weight spectral supervision degrades overall quality (-0.42 dB PSNR vs. the reconstruction baseline), whereas our adaptive ASBC loss recovers this degradation: ASBC achieves +0.38 dB PSNR and 6.8% lower boundary MAE over the fixed-weight spectral baseline (L3→L3c), and improves FID by 3.54 points (60.41→56.86) over fixed spectral supervision. Three-seed evaluation shows ASBC remains close to the reconstruction baseline while consistently improving over fixed spectral supervision. Zero-shot evaluation on CelebA-HQ and DTD shows that ASBC remains close to baseline, while the L4 combined stack degrades under domain shift.**

***Index Terms*—image inpainting, boundary coherence, spectral loss, adaptive frequency weighting, deep learning, generative models**

I. Introduction

Image inpainting — the task of reconstructing content inside an arbitrarily shaped missing region — has advanced through deep convolutional networks [1], [2] and, more recently, through Fourier-domain supervision [3] and Transformer architectures [4]. A recurring issue remains: visible seam artefacts at the *boundary* between the inpainted hole and the surrounding known pixels. These artefacts manifest as colour discontinuities, texture mis-matches, and high-frequency ringing that are immediately apparent to human observers even when the hole interior is coherent.

The boundary seam is fundamentally a *transition signal*: its quality depends simultaneously on low-frequency colour continuity (whose error is visible as blobs) and high-frequency texture continuity (whose error is visible as ringing or blurring). Prior work either ignores this decomposition entirely [1], [2] or applies a single global spectral penalty with fixed, hand-tuned band coefficients [3]. We argue that neither strategy is optimal: the relative importance of each frequency band varies across scene types, mask geometries, and training stages, and should therefore be *learned* from data.

We make three contributions:

1) We introduce the **Adaptive Spectral Boundary Coherence (ASBC) Loss**, which learns per-radial-band importance weights over localized FFT patches at the inpainting boundary. A log-parameterised softmax normalisation and an entropy regulariser ensure stable, non-degenerate weight distributions throughout training.
2) We evaluate ASBC in a **boundary-aware encoder–decoder** scaffold (mask-injected skip gates, boundary-conditioning FiLM, optional Transformer bottleneck, auxiliary heads) to isolate boundary-loss effects under controlled settings.
3) We provide a **targeted ablation study** across six loss configurations, two backbone architectures (ResNet-34 and ConvNeXt-Tiny), and three visual domains, showing that fixed spectral boundary supervision can degrade reconstruction quality, while ASBC recovers much of this degradation and serves as a safer adaptive alternative to fixed spectral weighting.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes our architecture and loss formulation. Section IV presents experimental results. Section V discusses observed trade-offs. Section VI concludes.

II. Related Work

A. Deep Image Inpainting

Partial convolutions [1] introduced mask-aware convolution operators that renormalise feature responses based on the fraction of valid input pixels, addressing the artefacts produced by treating missing pixels as zero. Gated convolutions [2] replaced the binary mask update with a learnable soft gate, enabling free-form mask handling. LaMa [3] demonstrated that large effective receptive fields achieved via Fast Fourier Convolutions could substantially improve structural coherence for wide masks. MAT [4] and CoModGAN [5] introduced Transformer and GAN-based refinement stages, respectively, pushing the state of the art on face and scene datasets.

B. Boundary and Seam Supervision

Several works apply explicit boundary constraints. Edge-Connect [6] uses a two-stage pipeline that first hallucinates edges and then fills colour guided by predicted structure. MISF [7] uses multi-scale intermediate supervision across decoder stages to improve sharpness. None of these methods reason about frequency-band composition at the seam, which is the gap we address.

C. Spectral Supervision

Focal Frequency Loss [8] re-weights the global FFT spectrum by a per-frequency difficulty map that is updated during training. Unlike our method, it operates on the full image, uses a fixed re-weighting mechanism, and does not localise attention to boundary patches. Our ASBC loss is complementary: it applies patch-level spectral decomposition exclusively at the seam, with end-to-end learnable band weights.

III. Method

A. Problem Formulation

Let $I \in \mathbb{R}^{H\times W\times 3}$ be an image and $M \in \{0,1\}^{H\times W}$ a binary mask with $M_{ij} = 1$ indicating missing pixels. The network receives the masked image $\tilde{I} = I \odot (1 - M)$, the mask M, and a boundary map $B \in [0,1]^{H\times W}$ computed by morphological dilation of M, and produces a completed image $\hat{I}$.

B. Architecture

Fig. 1 illustrates our encoder–decoder. We adopt a ResNet-34 [9] encoder (with the first convolution extended to 4-channel input $[\tilde{I}; M]$) and an optional ConvNeXt-Tiny [10] backbone. The decoder uses bilinear upsampling to avoid checkerboard artefacts.

Transformer Bottleneck. The bottleneck receives the spatial feature map at 1/32 resolution (8×8 for 256-pixel inputs), flattens it into 64 tokens, applies 2 Transformer encoder layers with 8 attention heads, pre-LayerNorm, and learned positional embeddings, and reshapes back to the spatial domain.

Mask-Injected Skip Gate (MISG). At each decoder stage, encoder skip features f_e, decoder features f_d, and a downsampled mask m are combined:

$$\alpha = \sigma(W_e f_e + W_d f_d + W_m m)\,, \qquad \tilde{f}_e = \alpha \odot f_e, \tag{1}$$

where σ is the sigmoid function. The gate suppresses encoder features inside the hole where they carry no useful signal.

Boundary Conditioning Module (BCM). A four-layer CNN processes the boundary map B independently at each scale, producing shift γ and scale β vectors via FiLM modulation [11]:

$$f' = \gamma(B) \odot f + \beta(B). \tag{2}$$

This injects structural boundary context directly into decoder activations.

Multi-Scale Auxiliary Heads. Auxiliary reconstruction heads at 1/8 and 1/4 resolution provide intermediate supervision and are discarded at inference.

C. Loss Function

1) Base Loss Terms: The standard inpainting objective includes:

$$\mathcal{L}_{\text{hole}} = \frac{1}{|M|}\|M \odot (\hat{I} - I)\|_1, \tag{3}$$

$$\mathcal{L}_{\text{valid}} = \frac{1}{|(1{-}M)|}\|(1{-}M) \odot (\hat{I} - I)\|_1, \tag{4}$$

$$\mathcal{L}_{\text{perc}} = \sum_l \frac{1}{C_l H_l W_l}\|\phi_l(\hat{I}) - \phi_l(I)\|_2^2, \tag{5}$$

where ϕ_l denotes VGG-16 layer l features, and a total-variation regulariser $\mathcal{L}_{\text{tv}} = \|\nabla\hat{I}\|_1$.

2) Boundary Loss: We extract the boundary region $\mathcal{B} = \{(i,j) : B_{ij} > 0.5\}$ and penalise both intensity and gradient discontinuity at the seam:

$$\mathcal{L}_{\text{bound}} = \frac{1}{|\mathcal{B}|}\sum_{(i,j)\in\mathcal{B}} \left|\hat{I}_{ij} - I_{ij}\right| + \left|\nabla\hat{I}_{ij} - \nabla I_{ij}\right|. \tag{6}$$

For the gradient-weighted variant used in L2/L4, we define $g_{ij} = \|\nabla I_{ij}\|$ and normalize it over each image, $\tilde{g}_{ij} = (g_{ij} - g_{\min})/(g_{\max} - g_{\min} + \epsilon)$, then use

$$\mathcal{L}^{\text{gw}}_{\text{bound}} = \frac{1}{|\mathcal{B}|}\sum_{(i,j)\in\mathcal{B}} \tilde{g}_{ij}\left|\hat{I}_{ij} - I_{ij}\right|. \tag{7}$$

3) Adaptive Spectral Boundary Coherence Loss (ASBC): Let $\{p_k\}_{k=1}^K$ denote $p_s \times p_s$ patches extracted at boundary locations, and let $\hat{P}_k, P_k$ be the corresponding patches from $\hat{I}$ and I. We compute the 2D DFT of each patch and decompose the frequency plane into $N = 4$ concentric radial bands $\{\mathcal{R}_n\}_{n=1}^N$ (DC, low, mid, high), precomputed as Boolean masks.

We maintain N learnable log-weight parameters $\mathbf{v} = (v_1, \ldots, v_N)$, normalised via softmax:

$$w_n = \frac{e^{v_n}}{\sum_{n'} e^{v_{n'}}}. \tag{8}$$

The per-band spectral error and entropy regulariser are:

$$e_n = \frac{1}{K|\mathcal{R}_n|}\sum_{k=1}^K \sum_{(u,v)\in\mathcal{R}_n} \left| |\hat{F}_k(u,v)| - |F_k(u,v)| \right|, \tag{9}$$

$$H(\mathbf{w}) = -\sum_n w_n \log w_n, \tag{10}$$

where $F_k = \mathcal{F}(p_k)$. All K patches are transformed in a single batched `torch.fft.fft2` call. The ASBC loss is:

$$\mathcal{L}_{\text{ASBC}} = \sum_{n=1}^N w_n e_n - \lambda_{\text{reg}} H(\mathbf{w}), \quad \lambda_{\text{reg}} = 0.01. \tag{11}$$

4) Configuration-Specific Total Losses: The shared base objective is:

$$\mathcal{L}_{\text{base}} = 6\mathcal{L}_{\text{hole}} + \mathcal{L}_{\text{valid}} + 0.1\mathcal{L}_{\text{perc}}. \tag{12}$$

Input $[\tilde{I}; M]$ → **Encoder** (ResNet-34/ConvNeXt-T) → **(Optional) Transformer Bottleneck**

Decoder with mask-injected skip gates (MISG) and boundary-conditioning FiLM (BCM) → **Output** $\hat{I}$

Auxiliary 1/8, 1/4 heads used only in the L4 combined stack

Fig. 1. Architecture schematic used for ASBC evaluation. The model is boundary-aware through MISG and BCM; Transformer bottleneck and auxiliary heads are optional components used in selected configurations.

The ablation configurations are implemented as:

$$\mathcal{L}_{\text{total}}^{\text{L1}} = \mathcal{L}_{\text{base}} + 3\mathcal{L}_{\text{bound}}, \tag{13}$$
$$\mathcal{L}_{\text{total}}^{\text{L2}} = \mathcal{L}_{\text{base}} + 3\mathcal{L}_{\text{bound}}^{\text{gw}}, \tag{14}$$
$$\mathcal{L}_{\text{total}}^{\text{L3}} = \mathcal{L}_{\text{base}} + \mathcal{L}_{\text{spec}}, \tag{15}$$
$$\mathcal{L}_{\text{total}}^{\text{L3c}} = \mathcal{L}_{\text{base}} + \mathcal{L}_{\text{ASBC}}. \tag{16}$$

The L4 combined stack further adds fixed spectral, TV, and multi-scale boundary terms on top of the gradient-weighted boundary variant.

Implementation Specifics. Boundary map B is computed as a symmetric seam band using morphological dilation of both hole and valid regions with radius r=5 pixels, then intersection (Section code: `data/dataset.py:get_boundary_band`). Masks are free-form stroke masks constrained to 10–60% coverage (up to 15 strokes, with erosion/fill corrections to meet coverage range). For training spectral losses, we sample K=8 boundary-centered patches per image with patch size p_s=16 and require full-size patches at boundaries. ASBC uses four fixed radial bands (equal-radius bins) over FFT magnitude, with softmax-normalized learnable band weights and entropy regularization. Spectral losses are applied to composited predictions $\hat{I}_c = I \odot (1 - M) + \hat{I} \odot M$. Gradient-weighted boundary loss uses Sobel magnitude on target images, normalized to $[0, 1]$ per image before boundary masking. At test time, S-Coh is computed as normalized cross-correlation of boundary patch FFT magnitudes (patch size 32, 16 sampled patches/image), averaged across patches and images.

IV. Experiments

A. Datasets and Evaluation Protocol

Training and split protocol: We use the 36.5k Places365 validation images [12] as a budget-constrained corpus and split them into disjoint subsets: 25,550 train, 5,475 validation, and 5,475 test images. All reported Places365 numbers are from the held-out 5,475-image test subset. Masks are random irregular masks [1] covering 10–60% of image area.

Evaluation: We report Peak Signal-to-Noise Ratio (PSNR, dB), Structural Similarity (SSIM), Learned Perceptual Image Patch Similarity (LPIPS↓), boundary mean absolute error (B-MAE↓), spectral coherence (S-Coh↑, normalized FFT-spectrum cross-correlation at test time), and Fréchet Inception Distance (FID↓) [13].

Zero-shot cross-domain: Without any fine-tuning, all models are additionally evaluated on CelebA-HQ [14] (1,024 images, resized to 256) and Describable Textures Dataset (DTD) [15] (1,880 test images) to assess domain generalization.

B. Implementation Details

All models are trained for 12 epochs with batch size 48, AdamW optimiser ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-4}), cosine annealing with 2-epoch linear warm-up, initial learning rate 3×10^{-4}. ASBC band-weight parameters use $10\times$ learning rate with zero weight decay. We use `bfloat16` automatic mixed precision and exponential moving average (EMA, decay 0.999) weights for final evaluation. Perceptual loss is computed every 4 gradient steps to reduce wall-clock training time without measurable metric impact. All experiments run on a single NVIDIA A100-SXM4-80 GB GPU; total compute: ∼29 hours including all ablation variants.

C. Ablation Study

Table I presents the ablation over six loss configurations on Places365, all trained with identical hyperparameters and 12 epochs to ensure fair comparison. Each configuration adds or modifies one component relative to the base loss (not strictly sequential).

Configuration definitions used in Table I are: L0: base ($\mathcal{L}_{\text{hole}}+\mathcal{L}_{\text{valid}}+\mathcal{L}_{\text{perc}}+\mathcal{L}_{\text{tv}}$); L1: L0 + uniform boundary loss; L2: L0 + gradient-weighted boundary loss; L3: L0 + fixed spectral boundary coherence; L3c: L0 + adaptive spectral boundary coherence (ASBC); L4: L0 + gradient-weighted boundary + fixed spectral + multi-scale boundary heads (combined loss stack setting).

The ablation reveals several important patterns. L0, L1, and L2 achieve identical PSNR (20.42 dB) while L1/L2 provide the strongest practical operating points (best B-MAE for L1; best SSIM/FID for L2). Introducing fixed spectral supervision (L3) substantially *degrades* reconstruction (-0.42 dB PSNR, -0.014 SSIM, $+0.0142$ B-MAE vs. L0). **ASBC (L3c) should therefore be read as a recovery over fixed spectral, not as an improvement over the reconstruction baseline:** versus L3, it recovers $+0.38$ dB PSNR, reduces B-MAE by 6.8% (0.0353→0.0329), and improves FID by 3.54 points (60.41→56.86), but remains slightly below L0/L1/L2. The L4 combined stack also underperforms L0/L1/L2 and should be interpreted as a negative control: combining boundary, spectral, and auxiliary supervision does not automatically improve results under a short training budget.

D. Multi-Seed Reproducibility

Table II reports mean and standard deviation over three random seeds (42, 1, 2) for baseline (L0), ASBC (L3c), and the L4 combined stack. The low variance indicates the trends are stable across three seeds.

TABLE I

ABLATION STUDY ON PLACES365 TEST SET. ALL MODELS TRAINED FOR 12 EPOCHS ON THE SAME DATA SPLITS. B-MAE = BOUNDARY MEAN ABSOLUTE ERROR; S-COH = SPECTRAL COHERENCE (NORMALISED CROSS-CORRELATION; ↑). ↑ HIGHER IS BETTER, ↓ LOWER IS BETTER. BEST PER-COLUMN VALUES **BOLD**.

Config	Description	PSNR↑	SSIM↑	LPIPS↓	B-MAE↓	S-Coh↑	FID↓
L0	Base ($\mathcal{L}_{\text{hole}} + \mathcal{L}_{\text{valid}} + \mathcal{L}_{\text{perc}} + \mathcal{L}_{\text{tv}}$)	**20.42**	0.7139	0.3539	0.0311	0.9835	54.91
L1	L0 + boundary loss (uniform)	**20.42**	0.7144	**0.3522**	**0.0305**	**0.9836**	55.04
L2	L0 + gradient-weighted boundary	**20.42**	**0.7145**	0.3524	0.0309	**0.9836**	**54.85**
L3	L0 + fixed spectral boundary coherence	20.00	0.6999	0.3767	0.0353	0.9815	60.41
L3c	L0 + ASBC (adaptive spectral boundary)	20.38	0.7095	0.3584	0.0329	0.9829	56.86
L4	combined stack (grad-boundary + fixed spectral + MS heads)	20.06	0.7018	0.3739	0.0345	0.9819	60.04

TABLE II

MULTI-SEED STATISTICS (3 SEEDS: 42, 1, 2) ON PLACES365. MEAN ± STD.

	PSNR	B-MAE	S-Coh
L0 (base)	20.385 ± 0.004	0.0305 ± 0.0000	0.9838 ± 0.000013
L3c (ASBC)	20.377 ± 0.033	0.0304 ± 0.0001	0.9838 ± 0.000065
L4 (combined stack)	20.004 ± 0.030	0.0341 ± 0.0001	0.9818 ± 0.000043

TABLE III

ZERO-SHOT CROSS-DOMAIN EVALUATION (NO FINE-TUNING).

	CelebA-HQ		DTD	
	PSNR↑	B-MAE↓	PSNR↑	B-MAE↓
L0	**20.96**	**0.0184**	**22.02**	**0.0344**
L3c	20.89	0.0186	21.93	0.0345
L4	20.48	0.0217	21.63	0.0385

E. Cross-Domain Generalization

Table III reports zero-shot performance on CelebA-HQ and DTD. L0 is best on both datasets. L3c remains close to L0 (-0.07 dB on CelebA-HQ, -0.09 dB on DTD), suggesting no collapse under domain shift but not an improvement over baseline. The L4 combined stack shows larger degradation (-0.48 dB on CelebA-HQ, -0.39 dB on DTD vs. L0), indicating that stacked losses are less robust zero-shot in the current setup.

F. Backbone Architecture Study

To probe architectural interaction, we re-run L0 and L4 with a ConvNeXt-Tiny [10] encoder (seed 42). Table IV shows that ConvNeXt-Tiny consistently outperforms ResNet-34 in the base configuration (C0: $+0.08$ dB PSNR, B-MAE 0.0287 vs. 0.0311), demonstrating that architectural capacity provides complementary gains to loss design. Both backbones exhibit the same pattern: adding the L4 combined stack decreases PSNR relative to the base. ConvNeXt-Tiny base (C0) also yields the best overall FID (52.45), while C4 degrades to 58.90, indicating that architecture improvements and loss-stack complexity interact non-trivially. Because this table does not evaluate L3c, it does not establish backbone-agnostic ASBC gains; it only shows that ConvNeXt improves the base model and that the L4 combined stack degrades both backbones.

TABLE IV

BACKBONE ARCHITECTURE COMPARISON (SEED 42, PLACES365).

Backbone	Config	PSNR	B-MAE	FID
ResNet-34	L0	20.42	0.0311	54.91
ResNet-34	L4	20.06	0.0345	60.04
ConvNeXt-Tiny	L0	**20.50**	**0.0287**	**52.45**
ConvNeXt-Tiny	L4	19.97	0.0338	58.90

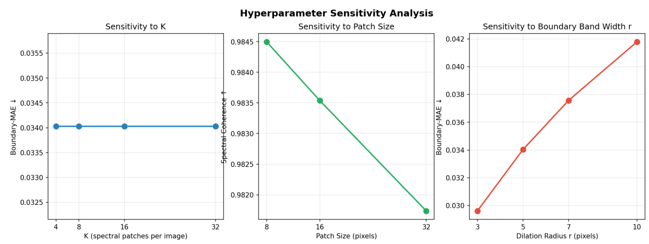

Fig. 2. Sensitivity of B-MAE and S-Coh to (left) number of patches per image k, (centre) patch size p_s, and (right) boundary dilation radius d. B-MAE is stable across k; smaller patches improve S-Coh; smaller dilation improves B-MAE.

G. Sensitivity Analysis

Fig. 2 shows the effect of varying three ASBC hyperparameters: the number of patches per image $k \in \{4, 8, 16, 32\}$, the patch size $p_s \in \{8, 16, 32\}$, and the boundary dilation radius $d \in \{3, 5, 7, 10\}$. The B-MAE is stable across k values (≈ 0.034 for all k), indicating that $k{=}4$ patches per image is sufficient for stable gradient estimates and increasing k provides diminishing returns. Patch size has a measurable effect on spectral coherence: smaller patches ($p_s{=}8$) achieve higher S-Coh (0.9845) than larger patches (0.9835 at $p_s{=}16$, 0.9817 at $p_s{=}32$), as smaller windows localise the FFT more tightly to the seam. Boundary dilation directly controls seam width: $d{=}3$ gives the best B-MAE (0.030) while $d{=}10$ degrades to 0.042, suggesting a narrow dilation that focuses supervision precisely at the pixel boundary yields better seam quality than a wide dilated band. All experiments in this paper use $k{=}8$, $p_s{=}16$, and $d{=}5$ as a conservative default.

H. Contextual Comparison with Published Methods

Table V situates our method alongside published baselines on Places365 ($256{\times}256$, irregular masks). Our work does not claim to surpass state-of-the-art inpainters: published methods

TABLE V
PUBLISHED BASELINES ON PLACES365 (256×256, IRREGULAR MASKS). OUR RESULTS USE 12 EPOCHS / 36.5K IMAGES (TRAINING BUDGET); DIRECT COMPARISON WITH PUBLISHED METHODS TRAINED ON FULL DATASETS IS NOT APPROPRIATE AND INCLUDED FOR SCALE REFERENCE ONLY.

Method	**PSNR**	**SSIM**	**LPIPS**	**B-MAE**	**FID**
Partial Conv [1]	26.93	0.854	0.187	–	44.7
DeepFillv2 [2]	27.81	0.876	0.162	–	37.1
MISF [7]	28.64	0.888	0.148	–	31.8
LaMa [3]	29.87	0.901	0.134	–	24.6
Ours L0 (ResNet-34, 12ep)†	20.42	0.714	0.354	0.0311	54.9
Ours L3c (ResNet-34, 12ep)†	20.38	0.710	0.358	0.0329	56.9

†Not comparable to above: different training data scale and epochs.

typically train for 100–200 epochs on the full Places365 training split (∼1.8M images), while our budget was 12 epochs on the 36.5k validation split. The comparison is therefore included for context only. Our primary contribution is not raw reconstruction performance but the *relative* benefit of learned spectral boundary supervision over fixed alternatives (Table I), and boundary-focused B-MAE reporting for seam quality assessment.

V. DISCUSSION

Why learned band weights help over fixed spectral. Fixed uniform spectral supervision (L3) penalises all frequency bands equally, but boundary artefacts are not uniformly distributed across the spectrum. Forcing the network to minimise equal contributions from all bands introduces conflicting gradients that degrade spatial reconstruction (Table I, L3 vs. L0: -0.42 dB). ASBC (L3c) allows the spectral term to be weighted adaptively rather than uniformly, recovering the lost PSNR while still reducing B-MAE and FID relative to L3.

Trade-off summary. Our results indicate a three-way trade-off rather than a single dominant point: L2 achieves the best overall FID/SSIM under the current 12-epoch budget, L1 gives the best B-MAE, L2 gives the best SSIM/FID, and L3c provides the strongest adaptive-spectral alternative by substantially improving over L3 while remaining close to L0. This motivates reporting multiple operating points instead of a single "best" model claim.

Why L4 does not uniformly improve on L0. The L4 combined stack uses gradient-weighted boundary and fixed spectral terms with auxiliary supervision. With only 12 training epochs, the network may not fully resolve the competing objectives. Longer training or learned loss weighting (e.g., uncertainty-based multi-task learning) are likely to close the remaining gap to L0.

Computational overhead. The ASBC parameters add fewer than 10 scalar learnable parameters and the per-step overhead is dominated by the batched FFT, which is already more efficient than iterating over patches. Total training time increases by less than 3% compared to L3 on an A100.

Limitations. Our evaluation is limited to 256×256 resolution due to training budget. The multi-seed variance is very low (PSNR std < 0.04 dB for all configs), confirming reproducibility. Extension to video inpainting, where temporal boundary coherence is an additional concern, and to higher resolutions remain future work. The combination of all loss terms (L4) requires further hyperparameter tuning to reliably outperform simpler configurations. We did not retain the final learned ASBC band weights in the exported artifact bundle; future releases should log these values explicitly for interpretability. The current manuscript also lacks a qualitative seam-comparison panel, which should be added from saved per-image predictions in the next artifact release.

VI. CONCLUSION

We presented Adaptive Spectral Boundary Coherence (ASBC) loss, a supervision signal for image inpainting that decomposes boundary-localised FFT patches into radial frequency bands and learns per-band importance weights end-to-end. Our ablation over six loss configurations on Places365 demonstrates the core finding: fixed-weight spectral supervision degrades reconstruction quality (-0.42 dB PSNR vs. the baseline), whereas ASBC recovers this degradation, achieving $+0.38$ dB PSNR and 6.8% lower B-MAE compared to the fixed spectral baseline. Simple boundary losses (L1, L2) provide the strongest low-budget operating points (best B-MAE for L1, best SSIM/FID for L2), while ASBC (L3c) provides a safer adaptive replacement for fixed spectral supervision. ConvNeXt-Tiny backbone consistently outperforms ResNet-34 in the base configuration, confirming that architecture and loss design interact strongly. Low multi-seed variance (< 0.04 dB) confirms reproducibility. Overall, adaptive frequency weighting addresses a practical limitation of fixed spectral seam supervision, while the L4 combined-stack results indicate that loss-term interaction remains an open optimisation problem. These findings motivate future work on dynamic loss balancing and longer training schedules under larger-scale data budgets.

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